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HACKNOVA'26 — SachVerify

PROJECT & PROBLEM STATEMENT DETAILS

Project / Solution Title: SachVerify + Beacon citizen app

Problem Statement Title: PS-2 · Disaster Information Verification Agent

GitHub Repository Link:
github.com/DevanshNarlawar06/Alebaple_hacknova (dev)

Live Prototype / Demo Link(Optional): Live demo on our laptop

Figma / Design Link (Optional): UI designs in repo: /design

TEAM INFORMATION

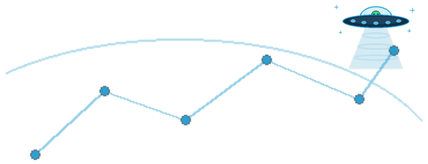
Team Name: Alebaple

Team Leader: Devansh Narlawar

Team Member 2: Akshat Patil

Team Member 3: Vedant Karne

Problem Statement & Existing Challenges



During floods, cyclones and earthquakes, claims spread faster than anyone can check them. **Responders can't tell which posts to trust.**

WHAT GOES WRONG (TYPICAL CASES)

Recycled media
An old flood video (e.g. Kolhapur 2019) is forwarded years later as "today".

Wrong place
A real alert for one district is attributed to another city.

Conflicting numbers
"300 mm rain", "dam burst", "10,000 evacuated" disagree with official figures.

Stale truth
A warning that expired hours ago is still being shared.

Language gap
Rumours spread in Hindi and Marathi; officials cross-check in English.

WHO IS AFFECTED

Responders & SDMA / DDMA control rooms
Triage hundreds of reports

Hospitals & NGOs
Decide where to send resources

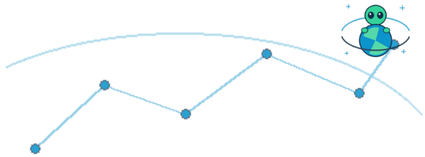
Citizens
Panic at fakes, or ignore real warnings

Why current tools fall short

- Official alerts (NDMA SACHET) are authoritative but don't answer rumours.
- Fact-checking is manual: hours to days.
- Platforms don't check a claim's place and time against official feeds.

Our assessment, from PS-2 context

Proposed Solution



SachVerify checks each disaster claim against official alerts, sensors, news, place and time, and gives one of four verdicts with the evidence behind it.

- **Verified**
  Trusted sources confirm it
- **Conflicting**
  Positive evidence contradicts it
- **Outdated**
  Was true, no longer current
- **Unverified**
  Not enough evidence either way

Absence of evidence is never treated as contradiction: no source found means Unverified, never "false".

Responder Console

PS-2 core

- Live story feed with status + confidence
- Evidence For / Against: source, time, distance, NLI score
- Map: official alert areas + claim pins
- Current vs Historical toggle; check-a-post box
- Replay demo streams posts live

Beacon citizen app

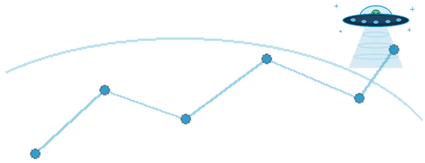
Beyond the brief

- Threat level for your location
- RED only on an official alert or a verified claim
- "Not true" rumour cards nearby
- Opt-in alarm sound; check a message
- "Why this alert?" shows the evidence

What makes it different

- Claims split into atomic facts; opinions excluded
- Each claim tied to one real event
- HISTORICAL vs CURRENT verdicts from validity windows
- Copied news stories count once
- Multilingual: a Hindi post checked against an English alert

Technical Approach & System Workflow



1 Ingest
Check-a-post / tipline, demo replay, live NDMA + GDACS polling

2 Extract
Gemini splits the post into atomic facts; grounding guard

3 Locate & bind
Geocode the place; tie the claim to one real event

4 Gather
Evidence by hazard from 5 source layers

5 Stance
Rules (numbers, physics, directives) + mDeBERTa NLI + e5

6 Validity
Validity windows give HISTORICAL vs CURRENT

7 Decide & explain
Rule table → status, confidence, cited evidence

Navy steps use AI models (Gemini, NLI). Every step is traced; the LLM only extracts and is grounding-checked.

DECISION RULES (FIRST MATCH WINS)

Conflicting	a fact is contradicted by positive evidence
Outdated	CURRENT check: true at claim time, since superseded
Verified	every factual proposition is supported
Unverified	otherwise: partial support, no coverage, place unresolved

Subjective statements ("worst quake ever") are shown but never scored.

Event binding (real pilot case)

"12 m tsunami at Acapulco", posted 20 Sep 2017, is tied to the **19 Sep Puebla M7.1** (USGS: inland, 236 km from Acapulco, no tsunami), not the 8 Sep Tehuantepec M8.2 that did cause one → **Conflicting**

Numeric conflict (synthetic demo)

"M6.8 quake hits Delhi" vs seismic record M3.1 → **Conflicting**

System Architecture

CLAIMS IN

- Check-a-post / tipline API
- Demo replay (labelled synthetic)
- Pilot claims (real 2017 events)

EVIDENCE · 5 LAYERS

1 Official
NDMA SACHET CAP, GDACS

2 Sensors
USGS, Open-Meteo

3 Local news
Google News (hi / mr / en), YouTube

4 National news
Google News, YouTube RSS

5 Other posts
independent posts, same story

+ Nominatim geocoding (cached)

Backend · FastAPI (Python 3.12)

Ingest poller
NDMA + GDACS every 3 min; typed connectors with cache

Event bus → SSE /api/stream
live verdicts, alerts, replay progress

Verifier agent · bounded state graph

extract → geocode → gather → bind event
stance → validity → decide → explain

deterministic steps · full trace stored per verdict

ML models
Gemini gemini-3-flash-preview (extraction)
mDeBERTa-v3 XNLI + multilingual-e5 (local)

Beacon threat engine
RED only on official alert or verified claim

SQLite (WAL) · schema v0.2
Extended additively from the earlier team's v0.1: claims, propositions, evidence + validity windows, versioned verdicts, eval runs

REST

SSE

Frontend

React + Vite + Tailwind + Leaflet

Console
responders: feed, evidence, map

Beacon
citizens: phone web app

Evaluation
metrics + confusion matrix

Planned (not built)

Recycled image/video checks
Mobile push notifications
Review UI + ~50-case benchmark
GPU acceleration

Technology Stack & Implementation

Frontend

React 18 + Vite 5
Console, Beacon and Eval pages

Tailwind CSS
shared design tokens (navy / teal / saffron)

Leaflet + OpenStreetMap
alert polygons, claim pins, quake points

Backend & API

FastAPI + Uvicorn + Pydantic
typed REST API, check-a-post / tipline endpoint

sse-starlette
live verdicts and alerts over SSE

httpx + feedparser
connectors with timeouts and response cache

AI / ML

Gemini gemini-3-flash-preview
atomic-fact extraction (JSON schema); fallbacks: lighter model, rules

mDeBERTa-v3 XNLI (multilingual)
stance: supports / contradicts

multilingual-e5-small
retrieval and copy detection

Data & geo

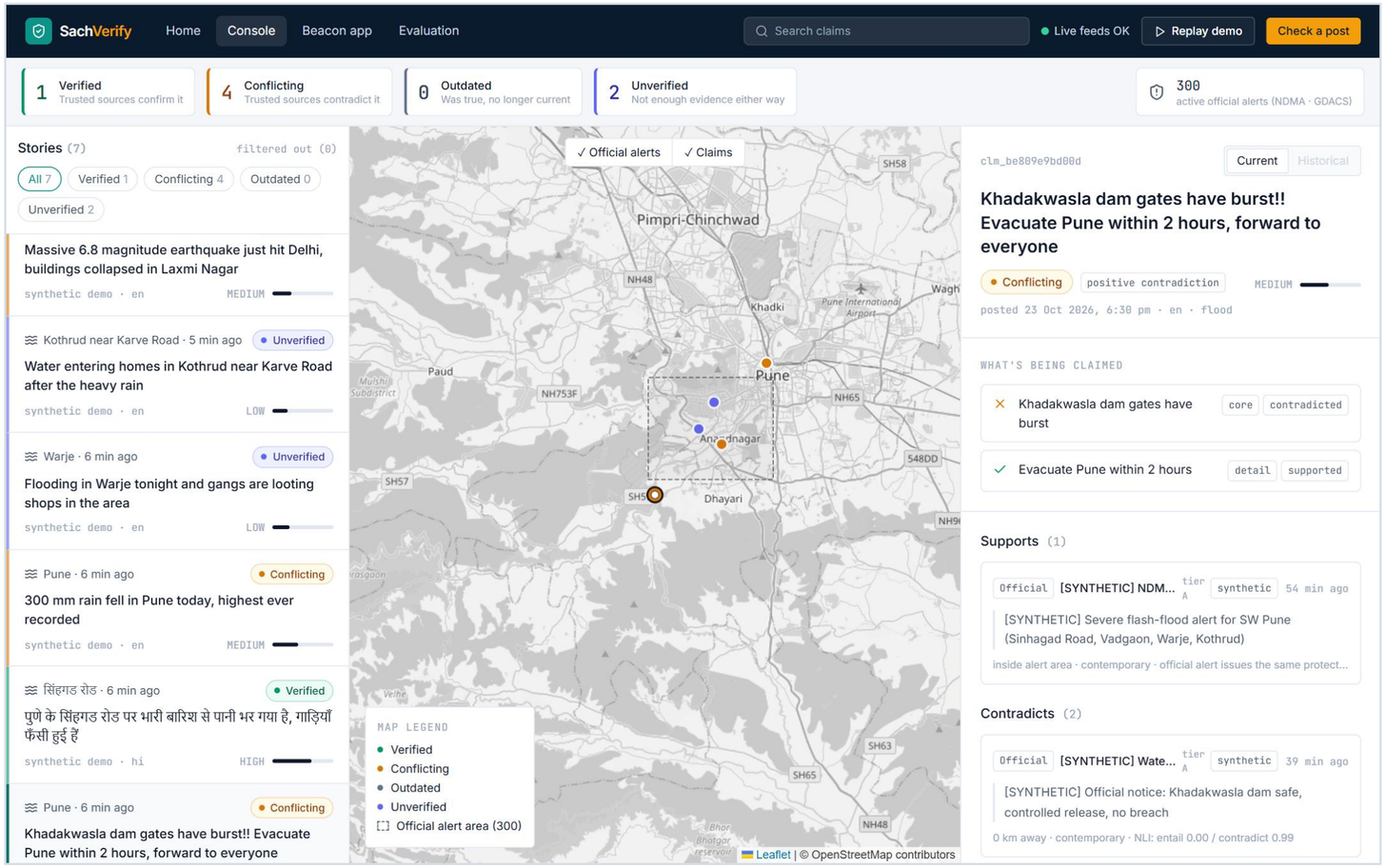
SQLite (WAL), schema v0.2
additive migration of the earlier team's dataset

shapely + Nominatim
point-in-alert-polygon checks; cached geocoding

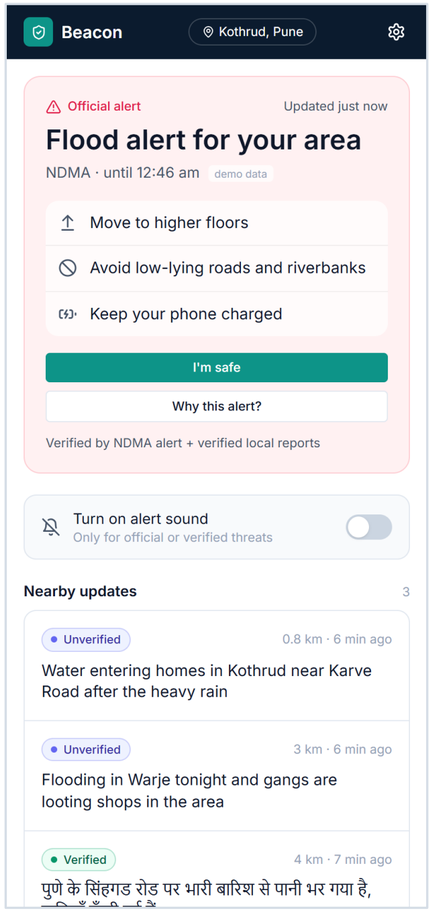
dateparser + langdetect
claimed times and languages (en / hi / mr)

Decisions: grounding guard (facts must appear in the post) · snapshot mode for an offline demo · every synthetic item labelled · keys only in .env · run_eval.py scores each run

Prototype & Key Features



Responder Console: a dam-burst rumour marked Conflicting, with the official notice that contradicts it



Beacon: official alert + rumour check

Working now

- Claim → verdict with cited evidence
- Map: 300 live NDMA / GDACS alert areas
- Current vs Historical toggle
- Hindi post verified against an English alert
- Beacon alert + rumour cards
- Replay demo over SSE

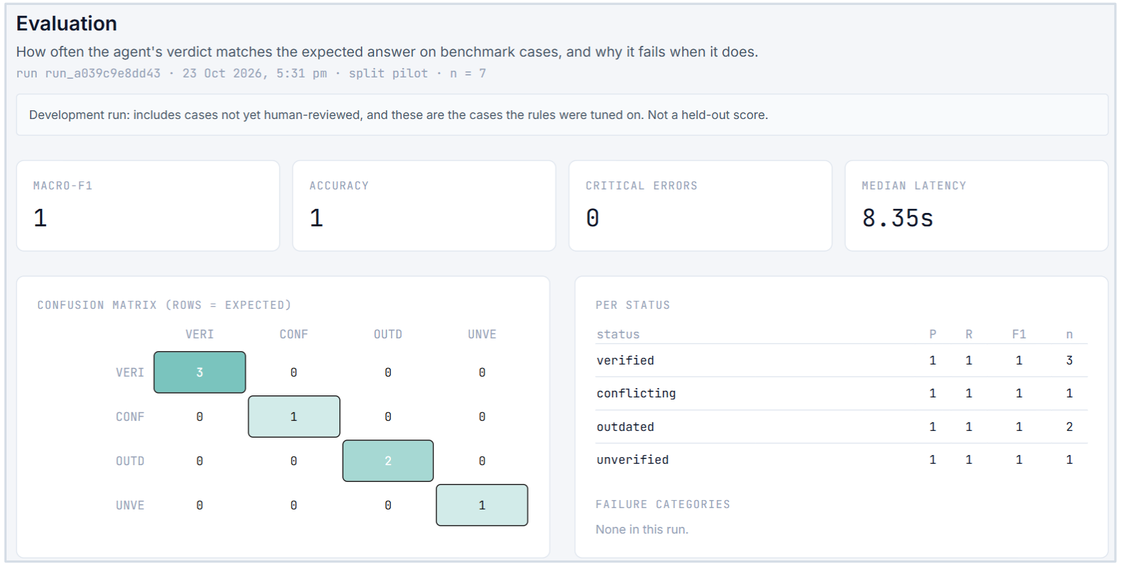
Data honesty

Pune flood and Delhi quake posts are synthetic and labelled in the UI.

Live NDMA / GDACS alerts are real.

Feasibility, Impact & Benefits

PILOT EVALUATION · APP SCREENSHOT



Evaluation

How often the agent's verdict matches the expected answer on benchmark cases, and why it fails when it does.

run run_a039c9e8dd43 · 23 Oct 2026, 5:31 pm · split pilot · n = 7

Development run: includes cases not yet human-reviewed, and these are the cases the rules were tuned on. Not a held-out score.

MACRO-F1	ACCURACY	CRITICAL ERRORS	MEDIAN LATENCY
1	1	0	8.35s

CONFUSION MATRIX (ROWS = EXPECTED)

	VERI	CONF	OUTD	UNVE
VERI	3	0	0	0
CONF	0	1	0	0
OUTD	0	0	2	0
UNVE	0	0	0	1

PER STATUS

status	P	R	F1	n
verified	1	1	1	3
conflicting	1	1	1	1
outdated	1	1	1	2
unverified	1	1	1	1

FAILURE CATEGORIES

None in this run.

Not a held-out accuracy. 7 / 7 correct, 0 critical errors, 8.35 s median on 5 pilot claims (7 checks). Rules were tuned on these cases; not yet human-reviewed.

BENEFITS

- **Responders:** triage with the evidence attached
- **Citizens:** alarms only for verified threats
- **Authorities:** see which rumours spread where

RISKS & MITIGATIONS

Risk	Mitigation
NLI misreads short or slangy posts	Numeric, geo and time rules come first; disagreement → Unverified
Live feeds down at the venue	Snapshot mode serves cached feeds
Local news early but wrong	Lower trust than official / sensor layers; conflicts shown side by side
Copied stories inflate confidence	Copies grouped; each group counts once
Gemini unavailable	Retry, lighter model, then rules-only extraction

FEASIBILITY

- **Technical:** one laptop; local NLI + embedding models; free public feeds, no paid APIs
- **Operational:** NDMA SACHET CAP is public and structured; snapshot mode works offline
- **Workflow:** control rooms already triage reports; the Console adds the evidence

Implemented vs Planned

PS-2 core: implemented

- Claim extraction (atomic facts)
- Source comparison, 5 layers
- Freshness via validity windows
- Contradiction detection
- Geographic verification
- Evidence For / Against per claim
- 4 statuses
- Dashboard + interactive map

Beyond the brief: implemented

- Beacon citizen app (verified-only alarm)
- Rumour-check cards
- HISTORICAL vs CURRENT verdicts
- Event binding (Puebla vs Tehuantepec)
- Copied stories count once
- Hindi / Marathi posts
- Live NDMA SACHET CAP polygons
- Offline snapshot mode

Planned: not built yet

- ~50-case human-reviewed benchmark
- Review UI for that benchmark
- Recycled image / video checks
- Mobile push notifications
- GPU acceleration

Why it matters

Evaluation covers 5 pilot claims so far. The reviewed benchmark comes before any accuracy claim.

Demo posts (Pune flood, Delhi quake) are synthetic and labelled; live NDMA / GDACS alerts are real.

THANK YOU!

Questions & Discussion

SachVerify · Team Alebaple · github.com/DevanshNarlawar06/Alebaple_hacknova